

A PYTHON BASED GUI For AM Modulation

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Amplitude modulation (AM) is a signal modulation technique that is widely used by radio stations for transmitting their programmes. This project proposes a Python GUI based simulator to study AM. A user can vary various parameters—such as modulation index, amplitude and frequency of modulating signal, and carrier signal—and observe the output. The project can be used to simulate the signal modulation by radio engineers for designing different carrier signals and their modulation and demodulation.

This GUI is developed in Python 3.10.1 by using Jupyter Notebook. As only two components are used for the project, as shown in the bill of material, there is no need of any circuit diagram. Just connect the LCD to Raspberry Pi via an HDMI cable connector.

For amplitude modulation, the carrier waveform is modulated according to instantaneous amplitude of the message (baseband) signal. To understand, let us consider a sinusoidal wave as the message signal that needs to be transmitted. Let us say, the amplitude of message signal is A_m and its frequency is f_m . The mathematical time-domain representation of the message signal would be:

$$m(t) = A_m \cos(2\pi f_m t) \quad \text{.....Eq 1}$$

The carrier signal is a high frequency radio wave with amplitude A_c and frequency f_c , which is represented by:

$$c(t) = A_c \cos(2\pi f_c t) \quad \text{.....Eq 2}$$

The equation of modulated

waveform is:

$$s(t) = A_c \left[1 + \left(\frac{A_m}{A_c} \right) \cos(2\pi f_m t) \right] \cos(2\pi f_c t) \quad \text{.....Eq 3}$$

The modulation index defines modulation level, which can be defined as:

$$\mu = \frac{A_m}{A_c} \quad (4)$$

Software installation

The software required for the project are Python 3.10.1, Tkinter, Matplotlib, and NumPy. First install the Anaconda Distribution or Python 3 environment in Raspberry Pi. Then install Tkinter, Matplotlib, and NumPy. Next, import these libraries

in code and prepare the code for signal modulator simulator.

Jupyter Notebook from Anaconda creates an interactive web based Python computing environment in any browser that is selected while installation.

Create a new notebook from the file menu of Jupyter IDE by selecting Python 3 as ipykernal. Rename the new notebook as Analog Communication Simulator.

This project uses functions from Tkinter, Matplotlib, and NumPy libraries, hence import their libraries. You can use pip install and conda install to install the libraries. For

example, *pip install matplotlib*

Next, create the graphical user interface (GUI) for the simulator. The GUI includes a window

BILL OF MATERIAL		
Components	Quantity	Description
Raspberry Pi	1	Raspberry Pi board to connect LCD
LCD display	1	For display
5V, 2A power adaptor	1	For power supply

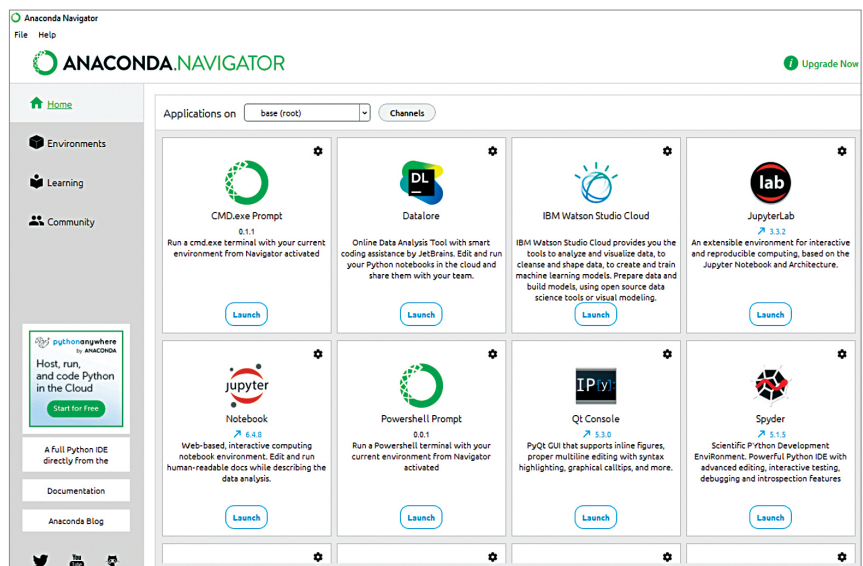


Fig. 1: Anaconda